SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

Creation Date 27-Oct-2015 **Revision Date** 28-Jan-2024 **Revision Number** 3

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description:	**alpha-Methylstyrene**
Cat No. :	**L03609**
Synonyms	Isopropenylbenzene; 2-Phenyl-1-propene
Index No	601-027-00-6
CAS No	98-83-9
EC No	202-705-0
Molecular Formula	C9 H10
REACH registration number	-

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Sector of use	SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites
Product category	PC21 - Laboratory chemicals
Process categories	PROC15 - Use as a laboratory reagent
Environmental release category	ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company .

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US:**001-201-796-7100 / **Europe:** +32 14 57 52 99
CHEMTREC Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Flammable liquids	Category 3 (H226)

Health hazards

Aspiration Toxicity	Category 1 (H304)
Serious Eye Damage/Eye Irritation	Category 2 (H319)
Skin Sensitization	Category 1 (H317)
Reproductive Toxicity	Category 2 (H361)
Specific target organ toxicity - (single exposure)	Category 3 (H335)

Environmental hazards

Chronic aquatic toxicity	Category 2 (H411)

Full text of Hazard Statements: see section 16

2.2. Label elements

Signal Word **Danger**

Hazard Statements

H226 - Flammable liquid and vapor
H304 - May be fatal if swallowed and enters airways
H317 - May cause an allergic skin reaction
H319 - Causes serious eye irritation
H335 - May cause respiratory irritation
H361 - Suspected of damaging fertility or the unborn child
H411 - Toxic to aquatic life with long lasting effects

Precautionary Statements

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician
P331 - Do NOT induce vomiting
P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention
P264 - Wash face, hands and any exposed skin thoroughly after handling
P337 + P313 - If eye irritation persists: Get medical advice/attention
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing
P280 - Wear protective gloves/protective clothing/eye protection/face protection
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)
Lachrymator (substance which increases the flow of tears)
Toxicity to Soil Dwelling Organisms
Contains a known or suspected endocrine disruptor
Contains a substance on the National Authorities Endocrine Disruptor Lists

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
.alpha.-Methylstyrene	98-83-9	EEC No. 202-705-0	>95	Eye Irrit. 2 (H319) STOT SE 3 (H335) Aquatic Chronic 2 (H411) Flam. Liq. 3 (H226) Asp. Tox. 1 (H304) Repr. 2 (H361) Skin Sens. 1 (H317)
1,2-Benzenediol, 4-(1,1-dimethylethyl)-	98-29-3	202-653-9	0.0015	Acute Tox. 4 (H302) Acute Tox. 4 (H312) Skin Corr. 1B (H314) Skin Sens. 1 (H317) Eye Dam. 1 (H318) Aquatic Acute 1 (H400) Aquatic Chronic 2 (H411)

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
.alpha.-Methylstyrene	STOT SE 3 (H335) :: C>=25%	-	-
1,2-Benzenediol, 4-(1,1-dimethylethyl)-	-	1	-

REACH registration number	-	
Components	**Reach Registration Number**	
alpha-Methylstyrene	01-2119472426-35	

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice If symptoms persist, call a physician.

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

Ingestion Clean mouth with water and drink afterwards plenty of water. Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.

Inhalation Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur. Risk of serious damage to the lungs (by aspiration).

Self-Protection of the First Aider Use personal protective equipment as required.

4.2. Most important symptoms and effects, both acute and delayed

May cause allergic skin reaction. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
Water spray, carbon dioxide (CO2), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

Extinguishing media which must not be used for safety reasons
No information available.

5.2. Special hazards arising from the substance or mixture

Flammable. Vapors may travel to source of ignition and flash back. Vapors may form explosive mixture with air. Containers may explode when heated. Vapors may form explosive mixtures with air.

Hazardous Combustion Products
Carbon monoxide (CO), Carbon dioxide (CO_2).

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system.

6.3. Methods and material for containment and cleaning up

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

Hygiene Measures
Handle in accordance with good industrial hygiene and safety practice.

7.2. Conditions for safe storage, including any incompatibilities

Keep away from heat, sparks and flame. Keep container tightly closed in a dry and well-ventilated place. Keep refrigerated.

Technical Rules for Hazardous Substances (TRGS) 510 Storage Class (LGK) (Germany) Class 3

Switzerland - Storage of hazardous substances Storage class - SC 3
https://www.kvu.ch/de/themen/stoffe-und-produkte
https://www.kvu.ch/fr/themes/substances-et-produits
https://www.kvu.ch/it/temi/sostanze-e-prodotti

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits
List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE -** 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

Component	European Union	The United Kingdom	France	Belgium	Spain
.alpha.-Methylstyrene	TWA: 50 ppm (8h) TWA: 246 mg/m³ (8h) STEL: 100 ppm (15min) STEL: 492 mg/m³ (15min)	STEL: 100 ppm 15 min STEL: 491 mg/m³ 15 min TWA: 50 ppm 8 hr TWA: 246 mg/m³ 8 hr	TWA / VME: 25 ppm (8 heures). indicative limit TWA / VME: 123 mg/m³ (8 heures). indicative limit STEL / VLCT: 100 ppm. indicative limit STEL / VLCT: 492 mg/m³. indicative limit Peau	TWA: 50 ppm 8 uren TWA: 246 mg/m³ 8 uren STEL: 100 ppm 15 minuten STEL: 492 mg/m³ 15 minuten	STEL / VLA-EC: 100 ppm (15 minutos). STEL / VLA-EC: 492 mg/m³ (15 minutos). TWA / VLA-ED: 50 ppm (8 horas) TWA / VLA-ED: 246 mg/m³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
.alpha.-Methylstyrene	TWA: 50 ppm 8 ore. Time Weighted Average TWA: 246 mg/m³ 8 ore. Time Weighted Average STEL: 100 ppm 15 minuti. Short-term STEL: 492 mg/m³ 15 minuti. Short-term	TWA: 50 ppm (8 Stunden). AGW - exposure factor 2 TWA: 250 mg/m³ (8 Stunden). AGW - exposure factor 2 TWA: 50 ppm (8 Stunden). MAK TWA: 250 mg/m³ (8 Stunden). MAK Höhepunkt: 100 ppm Höhepunkt: 500 mg/m³	STEL: 100 ppm 15 minutos STEL: 492 mg/m³ 15 minutos TWA: 50 ppm 8 horas TWA: 246 mg/m³ 8 horas	TWA: 20 mg/m³ 8 uren	TWA: 50 ppm 8 tunteina TWA: 250 mg/m³ 8 tunteina STEL: 100 ppm 15 minuutteina STEL: 490 mg/m³ 15 minuutteina

Component	Austria	Denmark	Switzerland	Poland	Norway
.alpha.-Methylstyrene	MAK-KZGW: 100 ppm 15 Minuten MAK-KZGW: 492 mg/m³ 15 Minuten MAK-TMW: 50 ppm 8 Stunden	TWA: 50 ppm 8 timer TWA: 246 mg/m³ 8 timer STEL: 492 mg/m³ 15 minutter STEL: 100 ppm 15 minutter	STEL: 100 ppm 15 Minuten STEL: 500 mg/m³ 15 Minuten TWA: 50 ppm 8 Stunden	STEL: 480 mg/m³ 15 minutach TWA: 240 mg/m³ 8 godzinach	TWA: 50 ppm 8 timer TWA: 240 mg/m³ 8 timer STEL: 75 ppm 15 minutter. value calculated STEL: 300 mg/m³ 15

	MAK-TMW: 246 mg/m³ 8 Stunden		TWA: 250 mg/m³ 8 Stunden		minutter. value calculated

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
.alpha.-Methylstyrene	TWA: 240 mg/m³ STEL : 485 mg/m³	TWA-GVI: 50 ppm 8 satima. TWA-GVI: 246 mg/m³ 8 satima. STEL-KGVI: 100 ppm 15 minutama. during the monitoring of exposure the relevant value of biological monitoring shall be taken into account as suggested by the Scientific Committee for Occupational Exposure Limits to Chemical Agents (SCOEL) STEL-KGVI: 492 mg/m³ 15 minutama.	TWA: 50 ppm 8 hr. TWA: 246 mg/m³ 8 hr. STEL: 492 mg/m³ 15 min STEL: 100 ppm 15 min	STEL: 100 ppm STEL: 492 mg/m³ TWA: 50 ppm TWA: 246 mg/m³	TWA: 250 mg/m³ 8 hodinách. Ceiling: 500 mg/m³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
.alpha.-Methylstyrene	TWA: 50 ppm 8 tundides. TWA: 246 mg/m³ 8 tundides. STEL: 100 ppm 15 minutites. STEL: 492 mg/m³ 15 minutites.	TWA: 50 ppm 8 hr TWA: 246 mg/m³ 8 hr STEL: 100 ppm 15 min STEL: 492 mg/m³ 15 min		STEL: 492 mg/m³ 15 percekben. CK TWA: 246 mg/m³ 8 órában. AK	STEL: 100 ppm STEL: 492 mg/m³ TWA: 50 ppm 8 klukkustundum. TWA: 240 mg/m³ 8 klukkustundum.

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
.alpha.-Methylstyrene	STEL: 100 ppm STEL: 492 mg/m³ TWA: 5 mg/m³ TWA: 50 ppm TWA: 246 mg/m³	TWA: 50 ppm IPRD TWA: 246 mg/m³ IPRD STEL: 100 ppm STEL: 492 mg/m³	TWA: 50 ppm 8 Stunden TWA: 246 mg/m³ 8 Stunden STEL: 100 ppm 15 Minuten STEL: 492 mg/m³ 15 Minuten	TWA: 50 ppm TWA: 246 mg/m³ STEL: 100 ppm 15 minuti STEL: 492 mg/m³ 15 minuti	TWA: 50 ppm 8 ore TWA: 246 mg/m³ 8 ore STEL: 100 ppm 15 minute STEL: 492 mg/m³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
.alpha.-Methylstyrene	MAC: 5 mg/m³	Ceiling: 492 mg/m³ TWA: 50 ppm TWA: 246 mg/m³	TWA: 50 ppm 8 urah TWA: 246 mg/m³ 8 urah STEL: 100 ppm 15 minutah STEL: 492 mg/m³ 15 minutah	Binding STEL: 100 ppm 15 minuter Binding STEL: 492 mg/m³ 15 minuter TLV: 20 ppm 8 timmar. NGV TLV: 98 mg/m³ 8 timmar. NGV	TWA: 50 ppm 8 saat TWA: 246 mg/m³ 8 saat STEL: 100 ppm 15 dakika STEL: 492 mg/m³ 15 dakika
1,2-Benzenediol, 4-(1,1-dimethylethyl)-	Skin notation MAC: 2 mg/m³				

Biological limit values
This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods
BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.
MDHS70 General methods for sampling airborne gases and vapours
MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography
MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas

chromatography

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)
Workers; See table for values

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
1,2-Benzenediol, 4-(1,1-dimethylethyl)- 98-29-3 (0.0015)				DNEL = 1.6mg/m^3

Predicted No Effect Concentration (PNEC)
See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
.alpha.-Methylstyrene 98-83-9 (>95)	PNEC = 0.008mg/L	PNEC = 0.583mg/kg sediment dw	PNEC = 0.01645mg/L	PNEC = 66.15mg/L	PNEC = 0.112mg/kg soil dw
1,2-Benzenediol, 4-(1,1-dimethylethyl)- 98-29-3 (0.0015)	PNEC = 1.2µg/L	PNEC = 6.9µg/kg sediment dw	PNEC = 1.2µg/L	PNEC = 0.16mg/L	PNEC = 0.68µg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
.alpha.-Methylstyrene 98-83-9 (>95)	PNEC = 0.0008mg/L	PNEC = 0.0583mg/kg sediment dw			
1,2-Benzenediol, 4-(1,1-dimethylethyl)- 98-29-3 (0.0015)	PNEC = 0.12µg/L	PNEC = 0.69µg/kg sediment dw			

8.2. Exposure controls

Engineering Measures
Use explosion-proof electrical/ventilating/lighting/equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection Goggles (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Nitrile rubber Neoprene Natural rubber PVC	See manufacturers recommendations	-	EN 374	(minimum requirement)

Skin and body protection Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection When workers are facing concentrations above the exposure limit they must use

	appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly
Large scale/emergency use	Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced **Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387
Small scale/Laboratory use	Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. **Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141 When RPE is used a face piece Fit Test should be conducted
Environmental exposure controls	Prevent product from entering drains. Do not allow material to contaminate ground water system.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance	Colorless	
Odor	aromatic	
Odor Threshold	No data available	
Melting Point/Range	-23 °C / -9.4 °F	
Softening Point	No data available	
Boiling Point/Range	165 - 169 °C / 329 - 336.2 °F	@ 760 mmHg
Flammability (liquid)	Flammable	On basis of test data
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	**Lower** 0.9 Vol% **Upper** 6.6 Vol%	
Flash Point	45 °C / 113 °F	**Method -** No information available
Autoignition Temperature	445 °C / 833 °F	
Decomposition Temperature	No data available	
pH	5-6	500 g/l aq.sol
Viscosity	0.94 cP at 20 °C	
Water Solubility	Insoluble	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	**log Pow**	
.alpha.-Methylstyrene	3.48	
1,2-Benzenediol, 4-(1,1-dimethylethyl)-	1.98	
Vapor Pressure	2.9 mbar @ 20 °C	
Density / Specific Gravity	0.909	
Bulk Density	Not applicable	Liquid
Vapor Density	4.1 (Air = 1.0)	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

Molecular Formula	C9 H10
Molecular Weight	118.18
Explosive Properties	explosive air/vapour mixtures possible

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under recommended storage conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization Hazardous polymerization may occur.
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid

Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.

10.5. Incompatible materials

Acids. Strong oxidizing agents. Finely powdered metals. Peroxides.

10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO_2).

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;
Oral Based on available data, the classification criteria are not met
Dermal Based on available data, the classification criteria are not met
Inhalation Based on available data, the classification criteria are not met

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
.alpha.-Methylstyrene	LD50 = 4900 mg/kg (Rat)	14560 mg/kg (Rabbit)	22.85 mg/L/6h (Rat)
1,2-Benzenediol, 4-(1,1-dimethylethyl)-	815 mg/kg (Rat)	1331 mg/kg (Rat)	-

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; Category 2

(d) respiratory or skin sensitization;
Respiratory No data available
Skin Category 1

May cause sensitization by skin contact

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

Component	EU	UK	Germany	IARC
.alpha.-Methylstyrene				Group 2B

(g) reproductive toxicity; Category 2
Reproductive Effects Contains ingredients that are suspected reproductive hazards.

(h) STOT-single exposure; Category 3

Results / Target organs Respiratory system.

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Category 1

Other Adverse Effects The toxicological properties have not been fully investigated.

Symptoms / effects,both acute and delayed Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

11.2. Information on other hazards

Endocrine Disrupting Properties .
Assess endocrine disrupting properties for human health Contains a substance on the National Authorities Endocrine Disruptor Lists

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity
Ecotoxicity effects Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

Component	Freshwater Fish	Water Flea	Freshwater Algae
.alpha.-Methylstyrene	LC50: 28 mg/L/48h (Leuciscus idus) LC50: 2,97 mg/L/96h (Brachydanio rerio)	EC50: 1,645 mg/L/48h	
1,2-Benzenediol, 4-(1,1-dimethylethyl)-	LC50 = 0.12 mg/L 96h	EC50=0.48 mg/L 48h	

Component	Microtox	M-Factor
1,2-Benzenediol, 4-(1,1-dimethylethyl)-		1

12.2. Persistence and degradability
Persistence Persistence is unlikely.
Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
.alpha.-Methylstyrene	3.48	15 - 140 dimensionless
1,2-Benzenediol, 4-(1,1-dimethylethyl)-	1.98	No data available

12.4. Mobility in soil Spillage unlikely to penetrate soil The product is insoluble and floats on water The product evaporates slowly . Is not likely mobile in the environment due its low water solubility. Spillage unlikely to penetrate soil

12.5. Results of PBT and vPvB assessment Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

12.6. Endocrine disrupting properties
Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

Assess endocrine disrupting properties for the environment	Contains a substance on the National Authorities Endocrine Disruptor Lists.

12.7. Other adverse effects

Persistent Organic Pollutant	This product does not contain any known or suspected substance
Ozone Depletion Potential	This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products	Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.
Contaminated Packaging	Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.
European Waste Catalogue (EWC)	According to the European Waste Catalog, Waste Codes are not product specific, but application specific.
Other Information	Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains.
Switzerland - Waste Ordinance	Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600 https://www.fedlex.admin.ch/eli/cc/2015/891/en

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number	UN2303
14.2. UN proper shipping name	ISOPROPENYLBENZENE
14.3. Transport hazard class(es)	3
14.4. Packing group	III

ADR

14.1. UN number	UN2303
14.2. UN proper shipping name	ISOPROPENYLBENZENE
14.3. Transport hazard class(es)	3
14.4. Packing group	III

IATA

14.1. UN number	UN2303
14.2. UN proper shipping name	ISOPROPENYLBENZENE
14.3. Transport hazard class(es)	3
14.4. Packing group	III

14.5. Environmental hazards	Dangerous for the environment Product is a marine pollutant according to the criteria set by IMDG/IMO
14.6. Special precautions for user	No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories
Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
.alpha.-Methylstyrene	98-83-9	202-705-0	-	-	X	X	KE-23939	X	X
1,2-Benzenediol, 4-(1,1-dimethylethyl)-	98-29-3	202-653-9	-	-	X	X	KE-11368	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
.alpha.-Methylstyrene	98-83-9	X	ACTIVE	X	-	X	X	X
1,2-Benzenediol, 4-(1,1-dimethylethyl)-	98-29-3	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed **KECL** - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
.alpha.-Methylstyrene	98-83-9	-	Use restricted. See item 75. (see link for restriction details)	-
1,2-Benzenediol, 4-(1,1-dimethylethyl)-	98-29-3	-	Use restricted. See item 75. (see link for restriction details)	-

REACH links
https://echa.europa.eu/substances-restricted-under-reach

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
.alpha.-Methylstyrene	98-83-9	Not applicable	Not applicable
1,2-Benzenediol, 4-(1,1-dimethylethyl)-	98-29-3	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals
Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?
Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at

work .
Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values
Take note of Directive 94/33/EC on the protection of young people at work
Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification See table for values

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
.alpha.-Methylstyrene	WGK2	
1,2-Benzenediol, 4-(1,1-dimethylethyl)-	WGK3	

Swiss Regulations
Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).
Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
.alpha.-Methylstyrene 98-83-9 (>95)	Prohibited and Restricted Substances		
1,2-Benzenediol, 4-(1,1-dimethylethyl)- 98-29-3 (0.0015)	Prohibited and Restricted Substances		

15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3
H304 - May be fatal if swallowed and enters airways
H317 - May cause an allergic skin reaction
H319 - Causes serious eye irritation
H335 - May cause respiratory irritation
H361 - Suspected of damaging fertility or the unborn child
H411 - Toxic to aquatic life with long lasting effects
H226 - Flammable liquid and vapor
H302 - Harmful if swallowed
H312 - Harmful in contact with skin
H314 - Causes severe skin burns and eye damage
H318 - Causes serious eye damage
H400 - Very toxic to aquatic life
H410 - Very toxic to aquatic life with long lasting effects

Legend

CAS - Chemical Abstracts Service
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
ENCS - Japanese Existing and New Chemical Substances
AICS - Australian Inventory of Chemical Substances
NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists
DNEL - Derived No Effect Level
RPE - Respiratory Protective Equipment
LC50 - Lethal Concentration 50%
NOEC - No Observed Effect Concentration
PBT - Persistent, Bioaccumulative, Toxic
TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
OECD - Organisation for Economic Co-operation and Development
BCF - Bioconcentration factor
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
ATE - Acute Toxicity Estimate
VOC - (volatile organic compound)

Key literature references and sources for data
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Training Advice
Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.
First aid for chemical exposure, including the use of eye wash and safety showers.

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This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

End of Safety Data Sheet